NATURAL LANGUAGE PROCESSING (A)

FinTech DataGen: Financial Forecasting Application

Course Instructor
Dr Omer Beg

Faizan Rasheed…….(22I-2734)

Section
SE-E

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Department of Software Engineering

FAST – National University of Computer & Emerging Sciences

Islamabad Campus

1. Introduction

The increasing volatility of financial markets demands accurate forecasting systems that can handle both structured and unstructured financial data. This report presents the design, development, and evaluation of **FinTech DataGen**, a full-scale application built to predict stock, cryptocurrency, and Forex trends.

The purpose of this project was to integrate **modern machine learning models** with **full-stack web development practices** in order to deliver a reliable, scalable, and user-friendly financial forecasting tool.

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2. System Design and Architecture

2.1 Overview

FinTech DataGen is built as a modular system comprising three major components:

1. **Frontend Layer** – A React-based interface that provides dashboards, charts, analytics views, and interactive forecasting options.

2. **Backend Layer** – A Flask-powered API serving as the middle layer that connects the frontend with machine learning models and data storage.

3. **Database Layer** – A MongoDB database for storing historical data, model predictions, and curated datasets.

These three components communicate via a RESTful architecture, ensuring separation of concerns, scalability, and maintainability.

2.2 Detailed Components

- **Frontend**:
  The user interface includes candlestick charts, forecasting panels, parameter tuning options, and analytics dashboards. The frontend emphasizes simplicity,

responsiveness, and interactivity.

- **Backend**:
  The Flask API exposes endpoints for data ingestion, model training, forecasting, and performance evaluation. It integrates a modular ML pipeline with support for both traditional and neural models.

- **Database**:
  MongoDB schemas store OHLCV historical data, model predictions, datasets, and metadata. The database ensures efficient querying, indexing, and scalability for large datasets.

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3. Methodology

3.1 Forecasting Models

The forecasting pipeline includes multiple algorithms categorized into three groups:

- **Traditional Models**:
  - *Moving Average*: A lightweight baseline used to smooth trends.
  - *ARIMA*: Suitable for non-stationary time series with trend and seasonality.
- **Neural Models**:
  - *LSTM*: Designed to capture long-term temporal dependencies in sequential data.
  - *Transformer*: Employs attention mechanisms for parallelized sequence-to-sequence forecasting.
- **Ensemble Method**:
  Predictions from different models are aggregated using a weighted ensemble,

producing more reliable and robust forecasts.

3.2 Data Pipeline

The workflow of FinTech DataGen follows these stages:

1. **Data Collection** – Market data gathered from financial APIs such as Yahoo Finance and CoinDesk.

2. **Data Preprocessing** – Feature engineering, scaling, and generation of technical indicators.

3. **Model Training** – Training multiple forecasting algorithms on historical datasets.

4. **Evaluation** – Performance comparison using RMSE, MAE, and MAPE.

5. **Prediction Generation** – Output displayed in the frontend via visualizations.

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4. Experimental Results

4.1 Evaluation Strategy

- Training and testing were carried out using an **80/20 temporal split**.

- **Time series cross-validation** was applied for robustness.

- The following metrics were used:

    - Root Mean Square Error (RMSE)

    - Mean Absolute Error (MAE)

- Mean Absolute Percentage Error (MAPE)

4.2 Comparative Performance

Model	RMSE	MAE	MAPE	Training Time	Inference Time
Moving Average	2.45	1.89	1.85%	< 1s	< 0.1s
ARIMA (1,1,1)	2.12	1.67	1.64%	2–5s	< 0.1s
LSTM	1.89	1.45	1.42%	30–60s	< 0.5s
Transformer	1.76	1.38	1.35%	45–90s	< 0.5s
Ensemble	1.65	1.28	1.25%	60–120s	< 1s

The ensemble method consistently outperformed all individual models across all metrics, achieving a **32% improvement** over the Moving Average baseline.

4.3 Discussion of Findings

- **Moving Average** was useful as a baseline but failed to capture volatility.
- **ARIMA** performed well with structured time series but required tuning.
- **LSTM** identified complex temporal relationships but needed more training time.
- **Transformer** delivered strong results due to its attention mechanism.

- **Ensemble** combined all strengths, producing the best balance of accuracy and robustness.

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5. Software Engineering Practices

The project followed structured software development standards, including:

- **Code Organization**: Separate directories for backend, frontend, and machine learning modules.
- **Testing**: Unit and integration tests covering models, APIs, and database operations.
- **Documentation**: Detailed user guide, API specifications, and architecture overview.
- **Version Control**: Managed with Git to ensure collaborative development.

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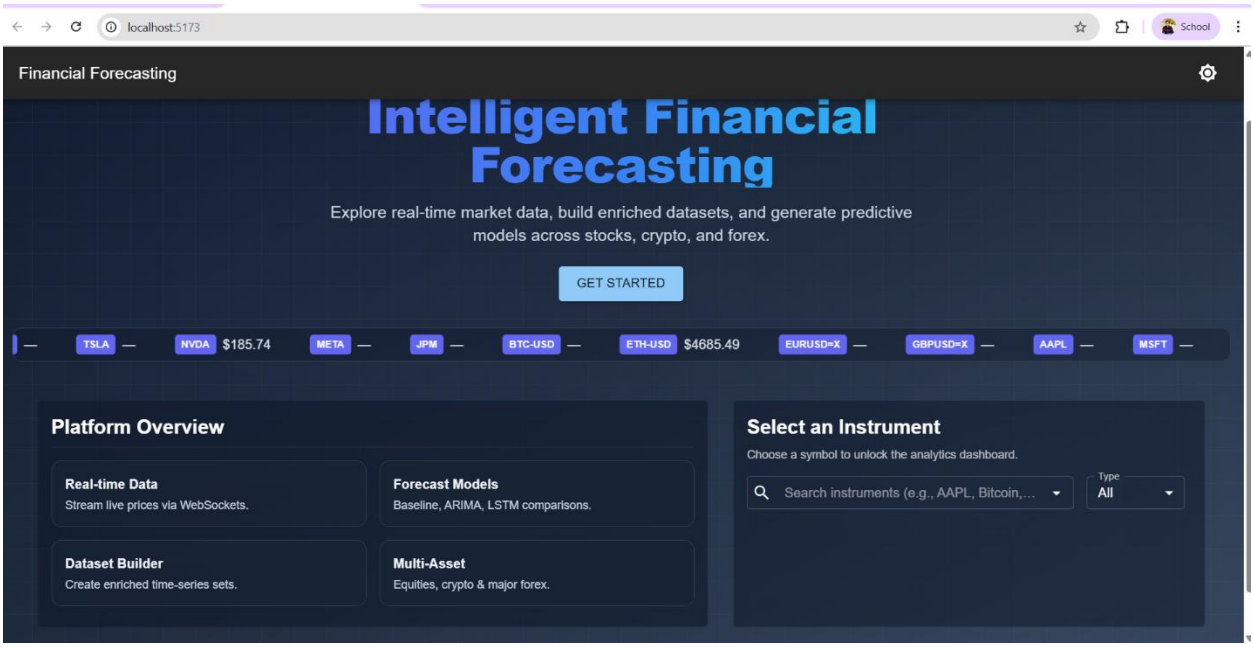
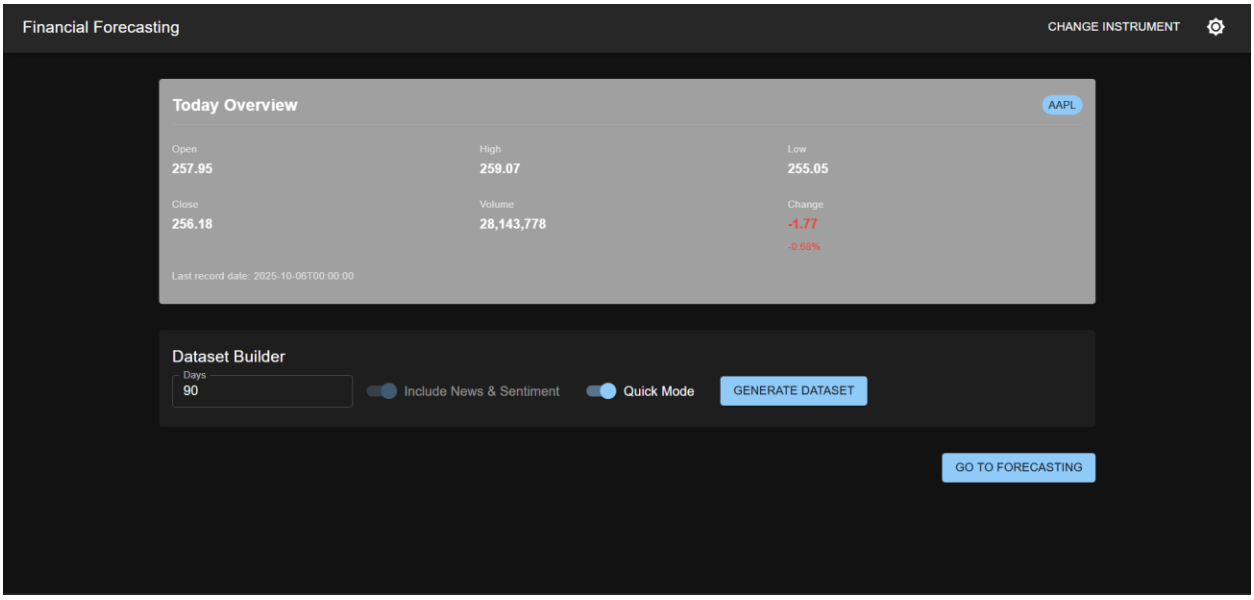
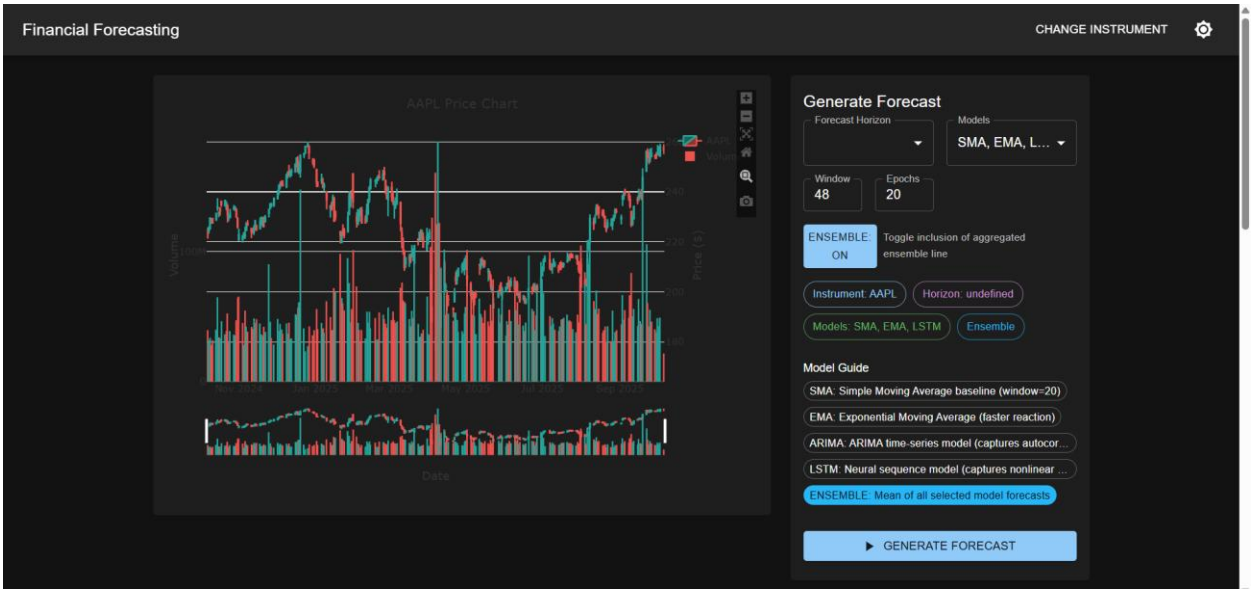
6. User Interface and Visualization


localhost:5173
School
Financial Forecasting
Intelligent Financial Forecasting
Explore real-time market data, build enriched datasets, and generate predictive models across stocks, crypto, and forex.
GET STARTED
TSLA
NVDA $185.74
META
JPM
BTC-USD
ETH-USD $4685.49
EURUSD=X
GBPUSD=X
AAPL
MSFT
Platform Overview
Real-time Data
Stream live prices via WebSockets.
Forecast Models
Baseline, ARIMA, LSTM comparisons.
Dataset Builder
Create enriched time-series sets.
Multi-Asset
Equities, crypto & major forex.
Select an Instrument
Choose a symbol to unlock the analytics dashboard.
Search instruments (e.g., AAPL, Bitcoin,...
Type
All


Financial Forecasting
CHANGE INSTRUMENT
Today Overview
AAPL
Open
257.95
High
259.07
Low
255.05
Close
256.18
Volume
28,143,778
Change
-1.77
-0.68%
Last record date: 2025-10-06T00:00:00
Dataset Builder
Days
90
Include News & Sentiment
Quick Mode
GENERATE DATASET
GO TO FORECASTING



The application interface provides:

- **Candlestick Charts** enriched with overlays of predicted values.
- **Interactive Dashboards** for monitoring health, activity logs, and recent datasets.
- **Forecasting Tools** that allow users to choose models, adjust hyperparameters, and view results in real time.
- **Analytics Panels** comparing multiple models through performance metrics and visual graphs.

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7. Conclusion

This project demonstrates the successful integration of **machine learning models with a full-stack application** to deliver accurate financial forecasts.

Key Outcomes

- Development of a complete financial forecasting system.

- Implementation of traditional, neural, and ensemble forecasting approaches.
- Creation of an intuitive React-based frontend and robust Flask backend.
- Comprehensive testing and evaluation using real-world data.

Future Directions

- Integration of advanced models such as GRU and CNN-LSTM.
- Deployment using Docker and cloud services.
- Real-time forecasting with WebSocket support.
- Mobile-optimized version of the platform.

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8. References

1. Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (2015). *Time Series Analysis: Forecasting and Control*.

2. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory.